VIZSGAREMEK

A **Baby-Rug Kft.** informatikai hálózata a vállalat mindennapi működésének alapját képezi, biztosítva a folyamatos és biztonságos adatáramlást a különböző részlegek között. Jelen dokumentáció célja a hálózati infrastruktúra felépítésének, működésének és karbantartási elveinek áttekintése. Részletesen bemutatja a hálózati eszközöket, a topológiát, a biztonsági megoldásokat, valamint az üzemeltetési irányelveket, amelyek hozzájárulnak a rendszer stabil és hatékony működéséhez. A dokumentum segítséget nyújt mind az új, mind a meglévő rendszergazdák számára a hálózat megértésében és kezelésében.

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2026

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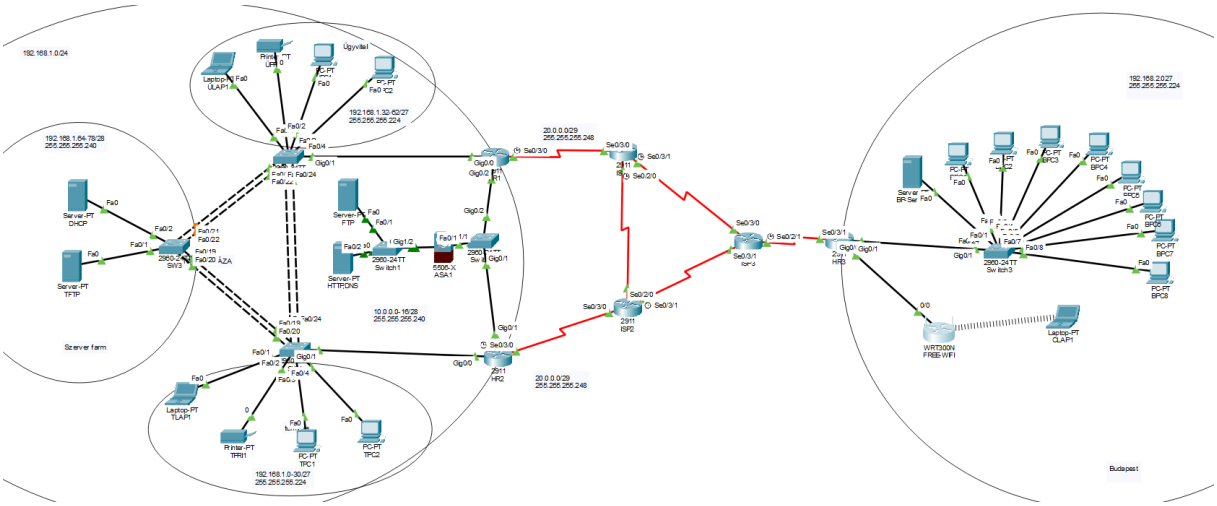


NYÍREGYHÁZI TELEPHELY

HÁLÓZATI TERV

NÉV	VLAN-ID	HÁLÓZAT CÍM	MASZK
TERMELÉS	10	192.168.1.0	255.255.255.224
SZERVERFARM	30	192.168.1.64	255.255.255.240
ÜGYVITEL	20	192.168.1.32	255.255.255.224
BELSŐHÁLÓ	77	192.168.1.80	255.255.255.240
DMZ	-----	192.168.2.0	255.255.255.248
OUTSIDE	-----	19.0.0.0	255.255.255.0

IPV4-ES CÍMTÁBLA

ESZKŐZ NÉV	INTERFACE	IP CÍM	MASZK	ALAPÉRTELMEZETT ÁTJÁRÓ
HR1	G0/0.10	192.168.1.3/27	255.255.255.224	-
	G0/0.20	192.168.1.35/27	255.255.255.224	-
	G0/0.30	192.168.1.65/28	255.255.255.240	-
	G0/0.77	192.168.1.81/28	255.255.255.240	-
HR2	G0/0.10	192.168.1.2/27	255.255.255.224	-
	G0/0.20	192.168.1.34/27	255.255.255.224	-
	G0/0.30	192.168.1.66/28	255.255.255.240	-
	G0/0.77	192.168.1.82/28	255.255.255.240	-
SW1	VALN77	192.168.1.83/28	255.255.255.240	192.168.1.81
SW2	VALN77	192.168.1.84/28	255.255.255.240	192.168.1.81
SW3	VALN77	192.168.1.85/28	255.255.255.240	192.168.1.81
TPRI1	VLAN10	192.168.1.5/27	255.255.255.224	192.168.1.1
TLAP1	VLAN10	DHCP		
TPC1	VLAN10	DHCP		
TPC2	VLAN10	DHCP		

ÜPRI1	VLAN20	192.168.1.36/27	255.255.255.224	192.168.1.33
ÜLAP1	VLAN20	DHCP		
ÜPC1	VLAN20	DHCP		
ÜPC2	VLAN20	DHCP		
DHCP	VLAN30	192.168.1.67/28	255.255.255.240	192.168.1.65
TFTP	VLAN30	192.168.1.69	255.255.255.240	192.168.1.65
HTTP, DNS	DMZ	10.0.0.3	255.255.255.240	10.0.0.1
ASA1	G1/1	19.0.0.0	255.255.255.0	-
	G1/2	10.0.0.1/29	255.255.255.248	-
FTP	DMZ	10.0.0.2/29	255.255.255.248	10.0.0.1

ISP

NÉV	HÁLÓZAT CÍM	MASZK
HR1 →ISP1	20.0.0.0	255.255.255.252
HR2→ISP2	21.0.0.0	255.255.255.252
ISP1→ISP2	22.0.0.0	255.255.255.252
ISP1→ISP3	23.0.0.0	255.255.255.252
ISP2→ISP3	24.0.0.0	255.255.255.252
ISP3→HR3	25.0.0.0	255.255.255.252

ISP			
HR1→ISP1	SE0/3/0	20.0.0.1/30	255.255.255.252
ISP1→HR1	SE0/3/0	20.0.0.2/30	255.255.255.252
HR2→ISP2	SE0/3/0	21.0.0.1/30	255.255.255.252
ISP2→HR2	SE0/3/0	21.0.0.2/30	255.255.255.252
ISP1→ISP2	SE0/2/0	22.0.0.1/30	255.255.255.252
ISP2→ISP1	SE0/2/0	22.0.0.2/30	255.255.255.252
ISP1→ISP3	SE0/3/1	23.0.0.1/30	255.255.255.252
ISP3→ISP1	SE0/3/0	23.0.0.2/30	255.255.255.252
ISP2→ISP3	SE0/3/1	24.0.0.1/30	255.255.255.252
ISP3→ISP2	SE0/3/1	24.0.0.2/30	255.255.255.252
ISP3→HR3	SE0/2/1	25.0.0.1/30	255.255.255.252
HR3→ISP3	SE0/3/1	25.0.0.2/30	255.255.255.252
HR1→HR3	tunnel1	192.168.100.1	255.255.255.252
HR3→HR1	tunnel1	192.168.100.2	255.255.255.252
HR3→HR2	runnel2	192.168.101.2	255.255.255.252
HR2→HR3	tunnel2	192.168.101.1	255.255.255.252

BUDAPEST

NÉV	HÁLÓZAT CÍM	MASZK
IRODA	192.168.2.0	255.255.255.224
HR3→VÁRÓ	192.168.3.0	255.255.255.252
VÁRÓ	192.168.50.0	255.255.255.128

BUDAPEST				
HR3	G0/0	192.168.2.0/27	255.255.255.224	-
	G0/1	192.168.3.0/24	255.255.255.0	-
BPC1-8	Fa0/1-8	DHCP		
BP-SERVER	Fa0/24	192.168.2.3	255.255.255.0	192.168.2.1
FREE-WIFI		192.168.4.0	255.255.255.0	

Helyszín	Eszköz	Interfész / VLAN	IPv6 cím	Prefix	Gateway
Nyíregyháza	HR1	G0/0.10	2001:db8:1:10::1	/64	-
		G0/0.20	2001:db8:1:20::1	/64	-
		G0/0.30	2001:db8:1:30::1	/64	-
		G0/0.77	2001:db8:1:77::1	/64	-
		SER0/3/0	2001:DB8:FF:1::1	/64	-
Nyíregyháza	HR2	G0/0.10	2001:db8:1:10::2	/64	-
		G0/0.20	2001:db8:1:20::2	/64	-
		G0/0.30	2001:db8:1:30::2	/64	-
		G0/0.77	2001:db8:1:77::2	/64	-
		SER0/3/0	2001:DB8:FF:2::1	/64	-
Nyíregyháza	TPRI1	VLAN10	2001:db8:1:10::10	/64	2001:db8:1:10::1
Nyíregyháza	TLAP1	VLAN10	DHCPv6	/64	2001:db8:1:10::1
Nyíregyháza	TPC1	VLAN10	DHCPv6	/64	2001:db8:1:10::1
Nyíregyháza	TPC2	VLAN10	DHCPv6	/64	2001:db8:1:10::1
Nyíregyháza	ÜPRI1	VLAN20	2001:db8:1:20::10	/64	2001:db8:1:20::1
Nyíregyháza	ÜLAP1	VLAN20	DHCPv6	/64	2001:db8:1:20::1
Nyíregyháza	ÜPC1	VLAN20	DHCPv6	/64	2001:db8:1:20::1
Nyíregyháza	ÜPC2	VLAN20	DHCPv6	/64	2001:db8:1:20::1
Nyíregyháza	DHCP/DNS	VLAN30	2001:db8:1:30::10	/64	2001:db8:1:30::1
Nyíregyháza	HTTPS	VLAN30	2001:db8:1:30::20	/64	2001:db8:1:30::1
Nyíregyháza	ASA1	G1/1 (Inside)	2001:db8:1:77::20	/64	-
Nyíregyháza	ASA1	G1/2 (DMZ)	2001:db8:1:100::1	/64	-
Nyíregyháza	FTP Server	DMZ	2001:db8:1:100::10	/64	2001:db8:1:100::1
WAN	ISP1	SER 0/3/0	2001:DB8:FF:1::2	/64	-
WAN	ISP1	SER0/2/0	2001:DB8:FF:3::1	/64	-

WAN	ISP1	SER0/3/1	2001:DB8:FF:4::1	/64	-
WAN	ISP2	SER0/2/0	2001:DB8:FF:3::2	/64	-
WAN	ISP2	SER0/3/0	2001:DB8:FF:2::2	/64	-
WAN	ISP2	SER0/3/1	2001:DB8:FF:5::1	/64	-
WAN	ISP3	SER0/2/1	2001:DB8:FF:6::2	/64	-
WAN	ISP3	SER0/3/0	2001:DB8:FF:4::2	/64	-
WAN	ISP3	SER0/3/1	2001:DB8:FF:5::2	/64	-
Budapest	HR3	G0/0	2001:db8:2:10::1	/64	-
Budapest	HR3	G0/1	2001:db8:2:20::1	/64	-
Budapest	HR3	SER0/3/1	2001:DB8:FF:6::1	/64	-
Budapest	BPC1-8	LAN	DHCPv6	/64	2001:db8:2:10::1
Budapest	BP-SERVER	LAN	2001:db8:2:10::10	/64	2001:db8:2:10::1

NYÍREGYHÁZI TELEPHELY:

Forgalom irányítás IPv4:

Alapértelmezett statikus útvonal:

ip route 0.0.0.0 0.0.0.0 20.0.0.2

OSPF dinamikus routing:

Egy dinamikus routing protokoll IPv4 hálózatokhoz. A legrövidebb útvonalat számolja ki a hálózatban (Dijkstra algoritmussal), és gyorsan reagál a változásokra.

router ospf 1
network 19.0.0.0 0.0.0.255 area 0
network 20.0.0.0 0.0.0.3 area 0

HR1:

```
HR1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 20.0.0.2 to network 0.0.0.0

     19.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       19.0.0.0/24 is directly connected, GigabitEthernet0/2
L       19.0.0.1/32 is directly connected, GigabitEthernet0/2
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/30 is directly connected, Serial0/3/0
L       20.0.0.1/32 is directly connected, Serial0/3/0
     21.0.0.0/30 is subnetted, 1 subnets
O       21.0.0.0/30 [110/65] via 19.0.0.2, 00:05:53, GigabitEthernet0/2
     22.0.0.0/30 is subnetted, 1 subnets
O       22.0.0.0/30 [110/128] via 20.0.0.2, 00:06:13, Serial0/3/0
     23.0.0.0/30 is subnetted, 1 subnets
O       23.0.0.0/30 [110/128] via 20.0.0.2, 00:06:13, Serial0/3/0
     24.0.0.0/30 is subnetted, 1 subnets
O       24.0.0.0/30 [110/129] via 19.0.0.2, 00:05:53, GigabitEthernet0/2
     25.0.0.0/30 is subnetted, 1 subnets
O       25.0.0.0/30 [110/192] via 20.0.0.2, 00:06:13, Serial0/3/0
     192.168.1.0/24 is variably subnetted, 8 subnets, 3 masks
C       192.168.1.0/27 is directly connected, GigabitEthernet0/0.10
L       192.168.1.3/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.32/27 is directly connected, GigabitEthernet0/0.20
L       192.168.1.35/32 is directly connected, GigabitEthernet0/0.20
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30
C       192.168.1.80/28 is directly connected, GigabitEthernet0/0.77
L       192.168.1.86/32 is directly connected, GigabitEthernet0/0.77
S    192.168.2.0/24 [1/0] via 192.168.100.2
     192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.100.0/30 is directly connected, Tunnel1
```

HR2:

```
router ospf 1
log-adjacency-changes
network 19.0.0.0 0.0.0.255 area 0
network 21.0.0.0 0.0.0.3 area 0
```

```
HR2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 21.0.0.2 to network 0.0.0.0

     19.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       19.0.0.0/24 is directly connected, GigabitEthernet0/1
L       19.0.0.2/32 is directly connected, GigabitEthernet0/1
     20.0.0.0/30 is subnetted, 1 subnets
O       20.0.0.0/30 [110/65] via 19.0.0.1, 00:09:39, GigabitEthernet0/1
     21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       21.0.0.0/30 is directly connected, Serial0/3/0
L       21.0.0.1/32 is directly connected, Serial0/3/0
     22.0.0.0/30 is subnetted, 1 subnets
O       22.0.0.0/30 [110/128] via 21.0.0.2, 00:10:09, Serial0/3/0
     23.0.0.0/30 is subnetted, 1 subnets
O       23.0.0.0/30 [110/129] via 19.0.0.1, 00:09:39, GigabitEthernet0/1
     24.0.0.0/30 is subnetted, 1 subnets
O       24.0.0.0/30 [110/128] via 21.0.0.2, 00:10:09, Serial0/3/0
     25.0.0.0/30 is subnetted, 1 subnets
O       25.0.0.0/30 [110/192] via 21.0.0.2, 00:09:59, Serial0/3/0
     192.168.1.0/24 is variably subnetted, 8 subnets, 3 masks
C       192.168.1.0/27 is directly connected, GigabitEthernet0/0.10
L       192.168.1.2/32 is directly connected, GigabitEthernet0/0.10
C       192.168.1.32/27 is directly connected, GigabitEthernet0/0.20
L       192.168.1.34/32 is directly connected, GigabitEthernet0/0.20
C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30
L       192.168.1.66/32 is directly connected, GigabitEthernet0/0.30
C       192.168.1.80/28 is directly connected, GigabitEthernet0/0.77
L       192.168.1.82/32 is directly connected, GigabitEthernet0/0.77
S    192.168.2.0/24 [1/0] via 192.168.101.2
     192.168.101.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.101.0/30 is directly connected, Tunnel2
L       192.168.101.1/32 is directly connected, Tunnel2
S*   0.0.0.0/0 [1/0] via 21.0.0.2
```

Forgalom irányítás IPv6:

OSPFv3 dinamikus routing:

Az OSPF IPv6-os változata. Ugyanúgy működik, mint az OSPFv4, csak IPv6 címeket kezel.

ipv6 router ospf 1

HR1:

```
HR1#show ipv6 route
IPv6 Routing Table - 19 entries
Codes: C - Connected, L - Local, S - Static, R - RIP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interar
       ND - ND Default, NDp - ND Prefix, DCE - Desti
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF e
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C   2001:DB8:1:10::/64 [0/0]
     via GigabitEthernet0/0.10, directly connected
L   2001:DB8:1:10::1/128 [0/0]
     via GigabitEthernet0/0.10, receive
C   2001:DB8:1:20::/64 [0/0]
     via GigabitEthernet0/0.20, directly connected
L   2001:DB8:1:20::1/128 [0/0]
     via GigabitEthernet0/0.20, receive
C   2001:DB8:1:30::/64 [0/0]
     via GigabitEthernet0/0.30, directly connected
L   2001:DB8:1:30::1/128 [0/0]
     via GigabitEthernet0/0.30, receive
C   2001:DB8:1:77::/64 [0/0]
     via GigabitEthernet0/0.77, directly connected
L   2001:DB8:1:77::1/128 [0/0]
     via GigabitEthernet0/0.77, receive
C   2001:DB8:FF:1::/64 [0/0]
     via Serial0/3/0, directly connected
L   2001:DB8:FF:1::1/128 [0/0]
     via Serial0/3/0, receive
O   2001:DB8:FF:2::/64 [110/192]
     via FE80::250:FFF:FE76:2801, Serial0/3/0
O   2001:DB8:FF:3::/64 [110/128]
     via FE80::250:FFF:FE76:2801, Serial0/3/0
O   2001:DB8:FF:4::/64 [110/128]
     via FE80::250:FFF:FE76:2801, Serial0/3/0
O   2001:DB8:FF:5::/64 [110/192]
     via FE80::250:FFF:FE76:2801, Serial0/3/0
O   2001:DB8:FF:6::/64 [110/192]
     via FE80::250:FFF:FE76:2801, Serial0/3/0
C   2001:DB8:FF:7::/64 [0/0]
     via GigabitEthernet0/2, directly connected
L   2001:DB8:FF:7::1/128 [0/0]
     via GigabitEthernet0/2, receive
O   2001:DB8:FF:8::/64 [110/193]
     via FE80::250:FFF:FE76:2801, Serial0/3/0
L   FF00::/8 [0/0]
     via Null0, receive
HR1#
```

HR2:

```
HR2#show ipv6 route
IPv6 Routing Table - 19 entries
Codes: C - Connected, L - Local, S - Static, R - R
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS inter
       ND - ND Default, NDp - ND Prefix, DCE - Des
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext
       D - EIGRP, EX - EIGRP external
C   2001:DB8:1:10::/64 [0/0]
     via GigabitEthernet0/0.10, directly connected
L   2001:DB8:1:10::2/128 [0/0]
     via GigabitEthernet0/0.10, receive
C   2001:DB8:1:20::/64 [0/0]
     via GigabitEthernet0/0.20, directly connected
L   2001:DB8:1:20::2/128 [0/0]
     via GigabitEthernet0/0.20, receive
C   2001:DB8:1:30::/64 [0/0]
     via GigabitEthernet0/0.30, directly connected
L   2001:DB8:1:30::2/128 [0/0]
     via GigabitEthernet0/0.30, receive
C   2001:DB8:1:77::/64 [0/0]
     via GigabitEthernet0/0.77, directly connected
L   2001:DB8:1:77::2/128 [0/0]
     via GigabitEthernet0/0.77, receive
O   2001:DB8:FF:1::/64 [110/192]
     via FE80::230:F2FF:FE8E:5701, Serial0/3/0
C   2001:DB8:FF:2::/64 [0/0]
     via Serial0/3/0, directly connected
L   2001:DB8:FF:2::1/128 [0/0]
     via Serial0/3/0, receive
O   2001:DB8:FF:3::/64 [110/128]
     via FE80::230:F2FF:FE8E:5701, Serial0/3/0
O   2001:DB8:FF:4::/64 [110/192]
     via FE80::230:F2FF:FE8E:5701, Serial0/3/0
O   2001:DB8:FF:5::/64 [110/128]
     via FE80::230:F2FF:FE8E:5701, Serial0/3/0
O   2001:DB8:FF:6::/64 [110/192]
     via FE80::230:F2FF:FE8E:5701, Serial0/3/0
C   2001:DB8:FF:7::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:FF:7::2/128 [0/0]
     via GigabitEthernet0/1, receive
O   2001:DB8:FF:8::/64 [110/193]
     via FE80::230:F2FF:FE8E:5701, Serial0/3/0
L   FF00::/8 [0/0]
     via Null0, receive
```

VPN,GRE tunnel:

HR1
interface Tunnel1
ip address 192.168.100.1
255.255.255.252
mtu 1476
tunnel source Serial0/3/0
tunnel destination 25.0.0.2

HR2
interface Tunnel2
ip address 192.168.101.1
255.255.255.252
mtu 1476
tunnel source Serial0/3/0
tunnel destination 25.0.0.2

Egy tunneling protokoll, amely lehetővé teszi különböző hálózati protokollok "becsomagolását" és átvitelét egy másik hálózaton keresztül (pl. interneten).

Biztonságos, titkosított kapcsolat két hálózat között nyilvános hálózaton (pl. interneten). Gyakran GRE-vel együtt használják.

```
C:\>tracert 192.168.2.10

Tracing route to 192.168.2.10 over a maximum of 3

  1   0 ms      0 ms      1 ms      192.168.1.3
  2   3 ms      *         5 ms      192.168.100.2
  3   2 ms      4 ms      2 ms      192.168.2.10

Trace complete.
```

```
tracert 192.168.2.10

cing route to 192.168.2.10 over a maximum of 3

   0 ms      0 ms      0 ms      192.168.1.2
   3 ms      3 ms      4 ms      192.168.101.2
   4 ms      3 ms      2 ms      192.168.2.10

ce complete.
```

3. rétegbeli redundancia megoldás -- HSRP

HSRP – Hot Standby Router Protocol

Cisco által fejlesztett redundancia protokoll. Több router közül egy az aktív (active), a többi készenlétben (standby), így ha az aktív kiesik, a standby átveszi a szerepét.

HR1:

```
interface GigabitEthernet0/0.77
 encapsulation dot1Q 77 native
 ip address 192.168.1.86 255.255.255.240
 ip nat inside
 ipv6 address 2001:DB8:1:77::1/64
 standby 77 ip 192.168.1.81
 standby 77 priority 110
 standby 77 preempt
```

HR2:

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.2 255.255.255.224
 ip helper-address 192.168.1.68
 ip nat inside
 ipv6 address FE80::10 link-local
 ipv6 address 2001:DB8:1:10::2/64
 ipv6 nd other-config-flag
 ipv6 dhcp server lan10
 standby 10 ip 192.168.1.1
```

```
HR1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig         10   110 P Active   local           192.168.1.2     192.168.1.1
Gig         20   110 P Active   local           192.168.1.34    192.168.1.33
Gig         30   110 P Active   local           192.168.1.66    192.168.1.65
Gig         77   110 P Active   local           192.168.1.82    192.168.1.81

HR2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig         10   100   Standby  192.168.1.3     local           192.168.1.1
Gig         20   100   Standby  192.168.1.35    local           192.168.1.33
Gig         30   100   Standby  192.168.1.65    local           192.168.1.65
Gig         77   100   Standby  192.168.1.86    local           192.168.1.81
```

Cisco fejlesztett redundancia protokoll, amelyben több router közül egy aktív szerepet tölt be, míg a többi készenléti állapotban van. Ha az aktív eszköz kiesik, egy standby router automatikusan átveszi a feladatát. Ez egy 3. rétegbeli redundancia megoldás

HR1

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.3 255.255.255.224
 ip helper-address 192.168.1.68
 ip nat inside
 ipv6 address FE80::10 link-local
 ipv6 address 2001:DB8:1:10::1/64
 ipv6 dhcp server lan10
 standby 10 ip 192.168.1.1
 standby 10 priority 110
 standby 10 preempt
```

HR2

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.2 255.255.255.224
 ip helper-address 192.168.1.68
 ip nat inside
 ipv6 address FE80::10 link-local
 ipv6 address 2001:DB8:1:10::2/64
 ipv6 nd other-config-flag
 ipv6 dhcp server lan10
 standby 10 ip 192.168.1.1
```

```
HR1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig         10   110 P Active   local           192.168.1.2     192.168.1.1
Gig         20   110 P Active   local           192.168.1.34    192.168.1.33
Gig         30   110 P Active   local           192.168.1.66    192.168.1.65
Gig         77   110 P Active   local           192.168.1.82    192.168.1.81
```

```
HR2#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig         10   100   Standby  192.168.1.3     local           192.168.1.1
Gig         20   100   Standby  192.168.1.35    local           192.168.1.33
Gig         30   100   Standby  192.168.1.65    local           192.168.1.65
Gig         77   100   Standby  192.168.1.86    local           192.168.1.81
```

Egyetlen publikus IP-címet több belső eszköz oszt meg különböző portok használatával (NAT overload néven is ismert).

HR1:

HR2:

```
interface Serial0/3/0
 ip address 20.0.0.1 255.255.255.252
 ip nat outside
 ipv6 address 2001:DB8:FF:1::1/64
 ipv6 ospf 1 area 0
 clock rate 128000
```

```
interface Serial0/3/0
 ip address 21.0.0.1 255.255.255.252
 ip nat outside
 ipv6 address 2001:DB8:FF:2::1/64
 ipv6 ospf 1 area 0
 clock rate 128000
```

```
C:\>tracert 192.168.2.13

Tracing route to 192.168.2.13 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.35
  2   *         3 ms      3 ms      192.168.101.2
  3   *         5 ms      3 ms      192.168.2.13
```

2. rétegbeli redundancia megoldás:

EtherChannel:

SW1
interface Port-channel1
switchport trunk native vlan 77
switchport trunk allowed vlan 10,20,30,77
switchport mode trunk

interface Port-channel2
switchport trunk native vlan 77
switchport trunk allowed vlan 10,20,30,77
switchport mode trunk

SW2
interface Port-channel1
switchport trunk native vlan 77
switchport trunk allowed vlan 10,20,30,77
switchport mode trunk

interface Port-channel3
switchport trunk native vlan 77
switchport trunk allowed vlan 10,20,30,77
switchport mode trunk

SW3
interface Port-channel1
switchport trunk native vlan 77
switchport trunk allowed vlan 10,20,30,77
switchport mode trunk

interface Port-channel3
switchport trunk native vlan 77
switchport trunk allowed vlan 10,20,30,77
switchport mode trunk

Több fizikai kapcsolat egy logikai kapcsolattá vonása. Növeli a sávszélességet és redundanciát biztosít.

SW1

```
Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------

1      Po1(SU)           LACP   Fa0/23(P) Fa0/24(P)
2      Po2(SU)           LACP   Fa0/19(P) Fa0/20(P)
```

SW3

```
Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------

2      Po2(SU)           LACP   Fa0/19(P) Fa0/20(P)
3      Po3(SU)           LACP   Fa0/21(P) Fa0/22(P)
```

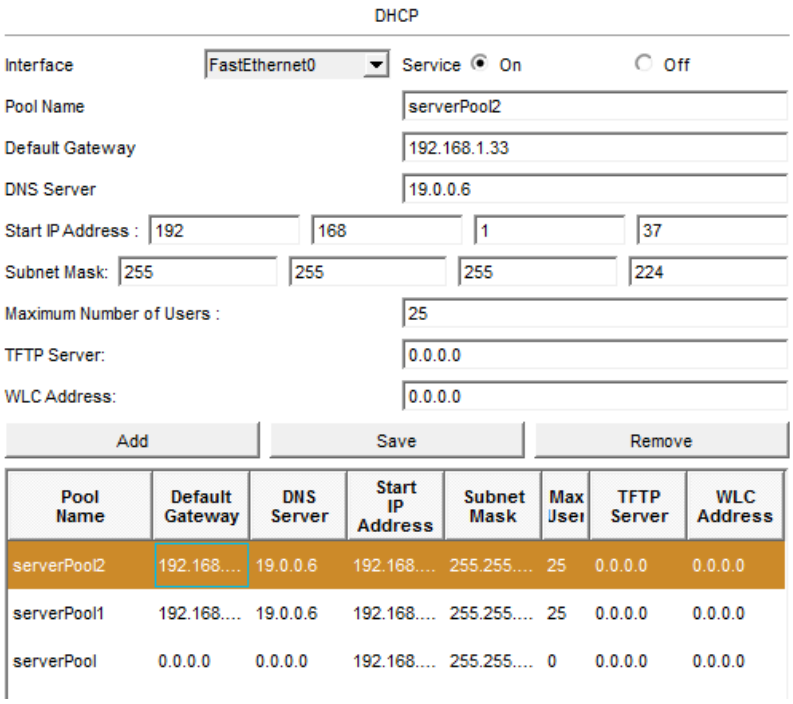
SW2

```
Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------

1      Po1(SU)           LACP   Fa0/23(P) Fa0/24(P)
3      Po3(SU)           LACP   Fa0/21(P) Fa0/22(P)
SW2#
```

IPv4 DHCP – Dynamic Host Configuration Protocol (IPv4):

Automatikusan kioszt IP-címeket és egyéb hálózati beállításokat (pl. gateway, DNS) IPv4 eszközöknek.

DHCP-server:

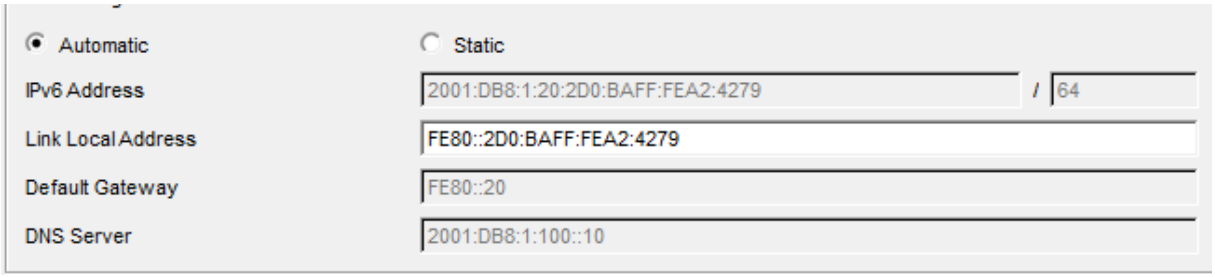


IPv6 DHCP – DHCPv6 (Dynamic Host Configuration Protocol for IPv6):

IPv6-os eszközöknek ad hálózati konfigurációt. Lehet állapotkövető (stateful) vagy állapotmentes (stateless).

HR1:

```
ipv6 dhcp pool lan10
 address prefix 2001:db8:1:10::/64 lifetime 172800 86400
 dns-server 2001:DB8:1:100::10
!
ipv6 dhcp pool lan20
 address prefix 2001:db8:1:20::/64 lifetime 172800 86400
 dns-server 2001:DB8:1:100::10
```



IPv4 NAT – Network Address Translation:

Privát IP-címeket fordít publikus IP-címekre, így a belső hálózat elérheti az internetet.

ASA- tűzfal:

```
object network dmz
 host 10.0.0.3
 nat (DMZ,outside) static 19.0.0.6
```

ACL – Access Control List:

Szabálylista, amely meghatározza, hogy milyen forgalom engedélyezett vagy tiltott egy hálózaton (IP, port, protokoll alapján).

```
access-list ping extended permit icmp any host 10.0.0.2
access-list ping extended permit icmp any host 10.0.0.3
access-list ping extended permit tcp any host 10.0.0.2 eq ftp
access-list ping extended permit udp any host 10.0.0.3 eq domain
access-list ping extended permit tcp any host 10.0.0.3 eq www
!
```

SSH – Secure Shell:

Biztonságos, titkosított távoli elérés eszközökhöz (pl. router, szerver). A Telnet biztonságos alternatívája.

HR1:

```
username admin password 0 cisco123

ip domain-name cisco.com

line con 0
password cisco123
line aux 0
line vty 0 4
password cisco123
login local
transport input ssh
line vty 5 15
password cisco123
login local
transport input ssh
```

```
C:\>ssh -l admin 192.168.1.3

Password:



HR1>ena
Password:
HR1#
```

Telnet – (Telecommunication Network):

Régi, nem titkosított távoli elérési protokoll.

SW1:

```
line vty 0 4
password cisco123
login
transport input telnet
line vty 5 15
password cisco123
login
transport input telnet
```

```
C:\>telnet 192.168.1.83
Trying 192.168.1.83 ...Open


User Access Verification

Password:
SW1>ena
Password:
SW1#
```

BUDAPESTI TELEPHELY:

IPV4 OSPF:

Egy dinamikus routing protokoll IPv4 hálózatokhoz. A legrövidebb útvonalat számolja ki a hálózatban (Dijkstra algoritmussal), és gyorsan reagál a változásokra.

router ospf 1
network 25.0.0.0 0.0.0.3 area 0

HR3:

```
HR3#show ip route o
     19.0.0.0/24 is subnetted, 1 subnets
O       19.0.0.0 [110/193] via 25.0.0.1, 02:14:11, Serial0/3/1
     20.0.0.0/30 is subnetted, 1 subnets
O       20.0.0.0 [110/192] via 25.0.0.1, 02:14:31, Serial0/3/1
     21.0.0.0/30 is subnetted, 1 subnets
O       21.0.0.0 [110/192] via 25.0.0.1, 02:14:31, Serial0/3/1
     22.0.0.0/30 is subnetted, 1 subnets
O       22.0.0.0 [110/192] via 25.0.0.1, 02:14:31, Serial0/3/1
     23.0.0.0/30 is subnetted, 1 subnets
O       23.0.0.0 [110/128] via 25.0.0.1, 02:14:31, Serial0/3/1
     24.0.0.0/30 is subnetted, 1 subnets
O       24.0.0.0 [110/128] via 25.0.0.1, 02:14:31, Serial0/3/1
```

IPV6 OSPFv3:

HR3:

ipv6 router ospf 1

ipv6 ospf 1 area 0

```
O   2001:DB8:FF:1::/64 [110/192]
     via FE80::2E0:A3FF:FEA0:201, Serial0/3/1
O   2001:DB8:FF:2::/64 [110/192]
     via FE80::2E0:A3FF:FEA0:201, Serial0/3/1
O   2001:DB8:FF:3::/64 [110/192]
     via FE80::2E0:A3FF:FEA0:201, Serial0/3/1
O   2001:DB8:FF:4::/64 [110/128]
     via FE80::2E0:A3FF:FEA0:201, Serial0/3/1
O   2001:DB8:FF:5::/64 [110/128]
     via FE80::2E0:A3FF:FEA0:201, Serial0/3/1
O   2001:DB8:FF:7::/64 [110/193]
     via FE80::2E0:A3FF:FEA0:201, Serial0/3/1
```

VPN, GRE TUNNEL:

Egy tunneling protokoll, amely lehetővé teszi különböző hálózati protokollok "becsomagolását" és átvitelét egy másik hálózaton keresztül (pl. interneten).

Biztonságos, titkosított kapcsolat két hálózat között nyilvános hálózaton (pl. interneten). Gyakran GRE-vel együtt használják.

interface Tunnel1
ip address 192.168.100.2 255.255.255.252
mtu 1476
tunnel source Serial0/3/1
tunnel destination 20.0.0.1

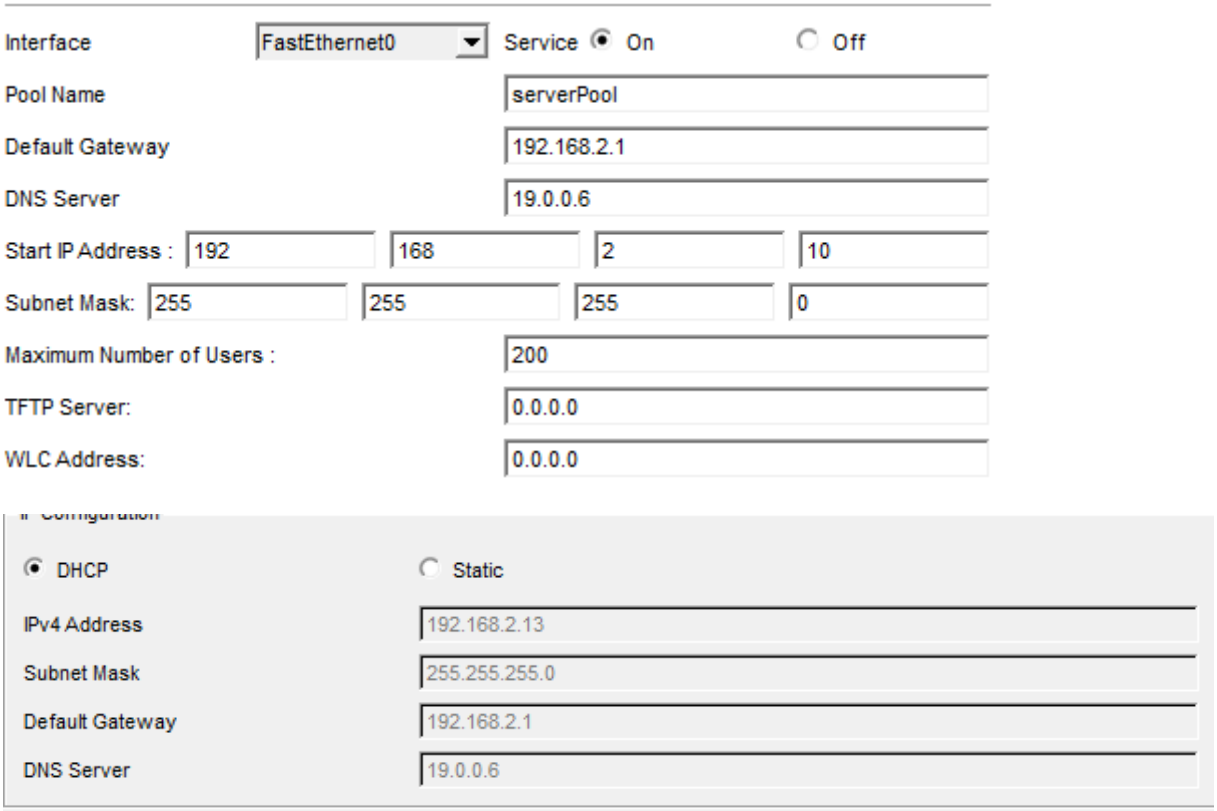
interface Tunnel2
ip address 192.168.101.2 255.255.255.252
mtu 1476
tunnel source Serial0/3/1
tunnel destination 21.0.0.1

```
C:\>tracert 192.168.1.8

Tracing route to 192.168.1.8 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.2.1
  2   22 ms     3 ms      3 ms      192.168.100.1
  3   *         3 ms      2 ms      192.168.1.8
```

IPv4 dhcp:

Automatikusan kioszt IP-címeket és egyéb hálózati beállításokat (pl. gateway, DNS) IPv4 eszközöknek.

BP-server:

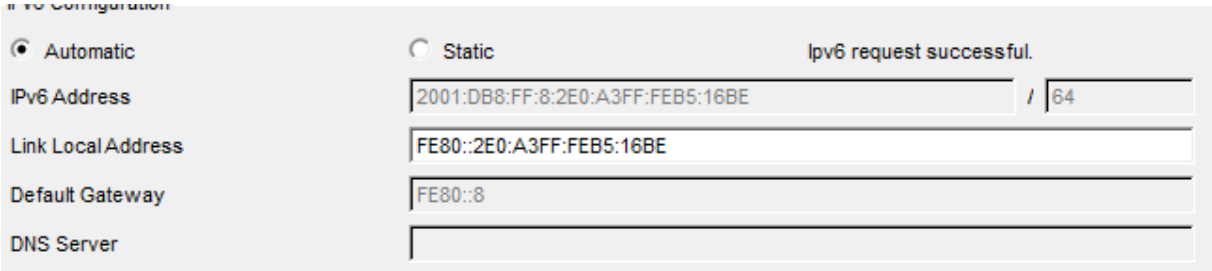


IPv6 DHCP:

IPv6-os eszközöknek ad hálózati konfigurációt. Lehet állapotkövető (stateful) vagy állapotmentes (stateless).

HR3:

ipv6 dhcp pool BP
address prefix 2001:db8:ff:4::/64 lifetime 172800 86400
dns-server 2001:DB8:FF:8::10



IPv4 NAT:

HR3:

Privát IP-címeket fordít publikus IP-címekre, így a belső hálózat elérheti az internetet.

ip nat inside source list 2 interface Serial0/3/1 overload

```
C:\>tracert 192.168.2.10

Tracing route to 192.168.2.10 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.3
  2   3 ms      *         5 ms      192.168.100.2
  3   3 ms      4 ms      3 ms      192.168.2.10

Trace complete.
```

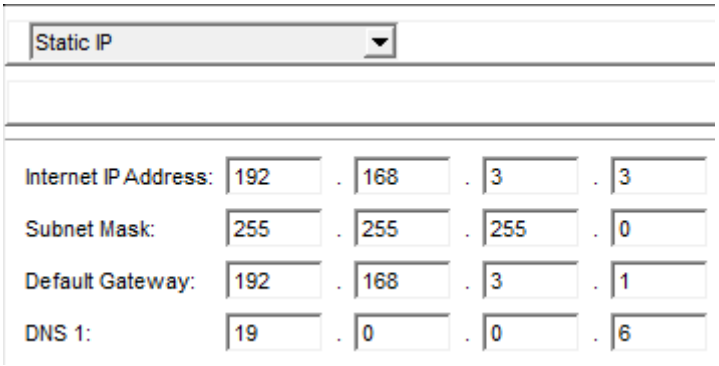
ACL – Access control list:

HR3:

Szabálylista, amely meghatározza, hogy milyen forgalom engedélyezett vagy tiltott egy hálózaton (IP, port, protokoll alapján).

```
access-list 2 permit 192.168.2.0 0.0.0.255
access-list 2 permit 192.168.3.0 0.0.0.255
```

FREE WIFI:



IP Address: 192 . 168 . 4 . 1

Subnet Mask: 255.255.255.0

DHCP Server: Enabled Disabled DHCP Reservation

Start IP Address: 192.168.4. 10

Maximum number of Users: 50

IP Address Range: 192.168.4. 10 - 59

Network Mode: Mixed

Network Name (SSID): visitor

Radio Band: Auto

Wide Channel: Auto

Standard Channel: 1 - 2.412GHz

SSID Broadcast: Enabled Disabled

Security Mode: WPA2 Personal

Encryption: AES

Passphrase: cisco123

Key Renewal: 3600 seconds

BPDU Guard – Bridge Protocol Data Unit Guard

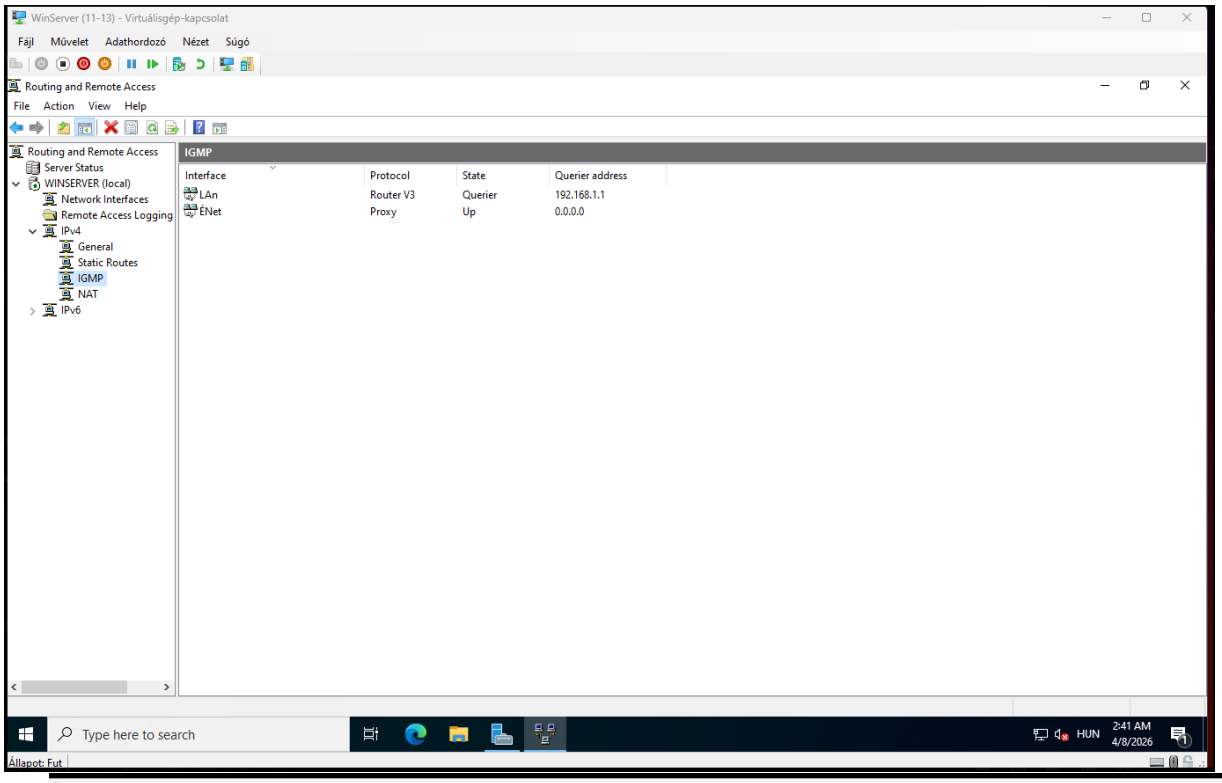
STP biztonsági funkció. Ha egy porton BPDU érkezik, ahol nem kellene (pl. végponti port), akkor letiltja azt.

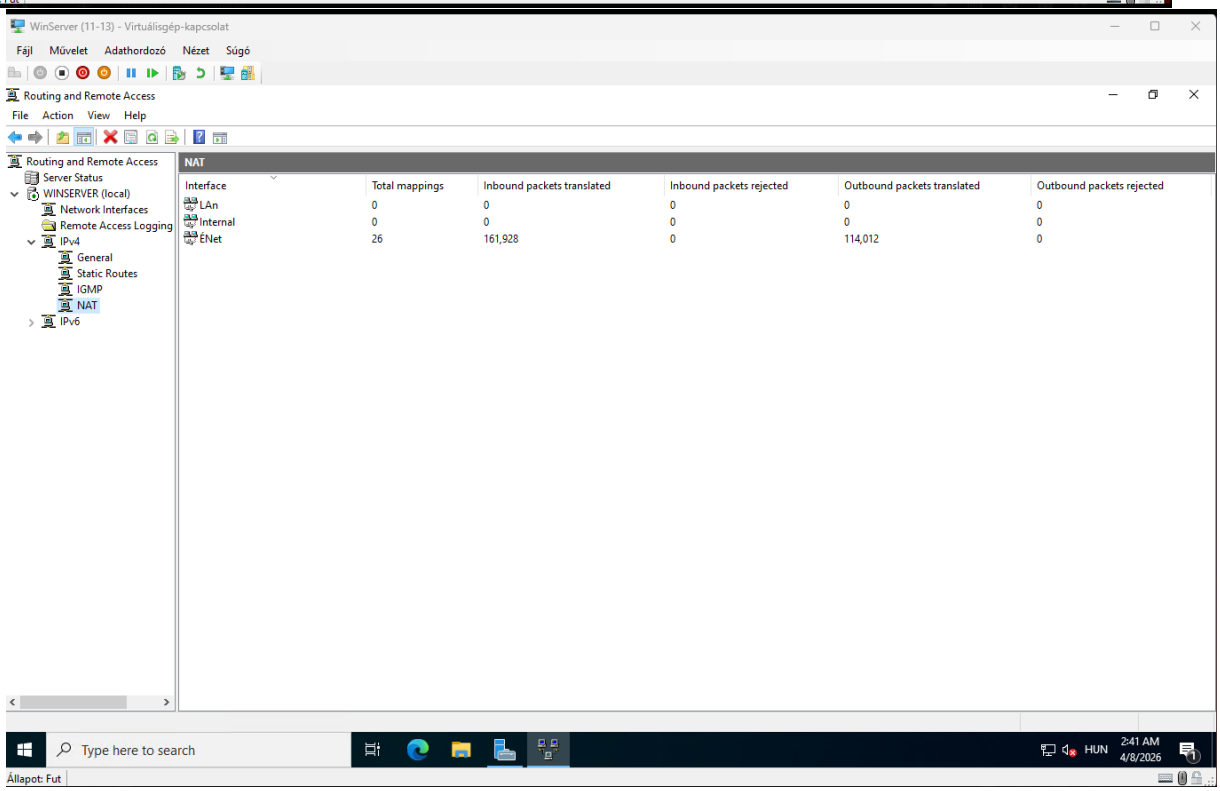
PortFast – (STP PortFast Feature)

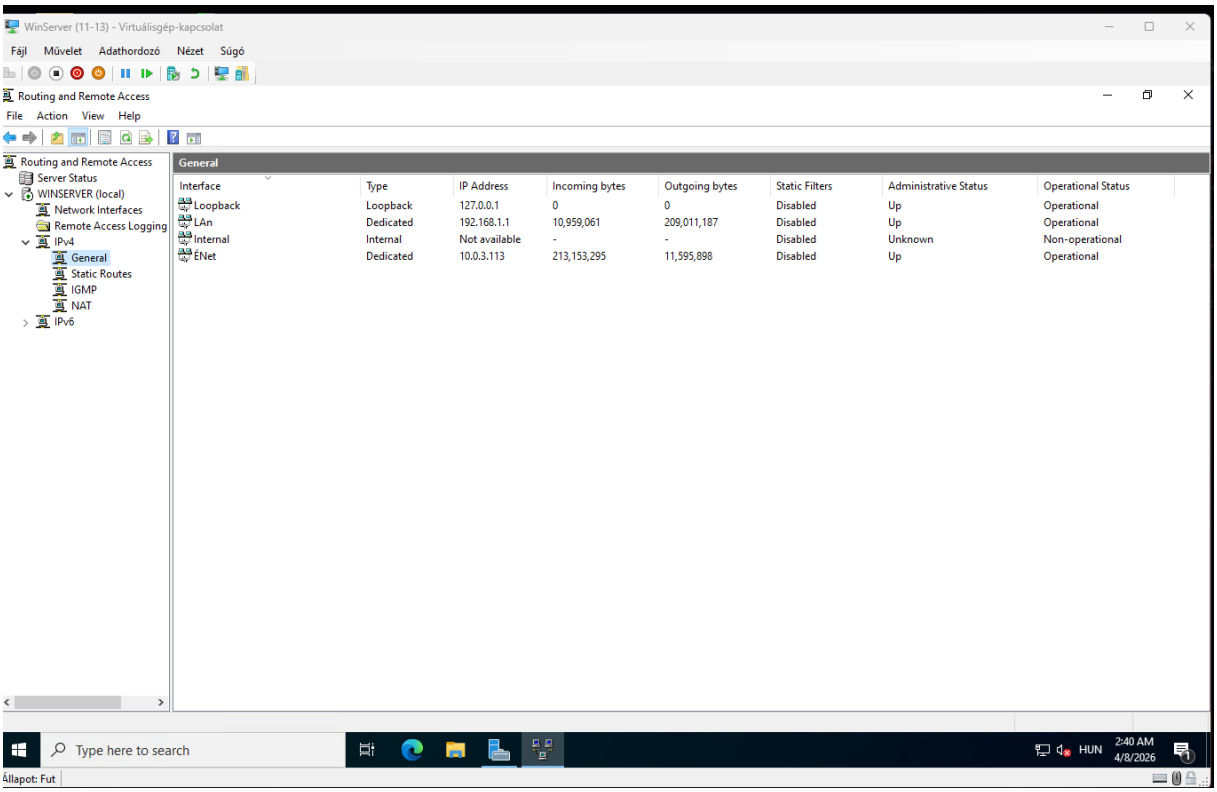
Gyorsítja a portok aktiválását végponti eszközök (pl. PC) esetén, kihagyva az STP tanulási fázisait.

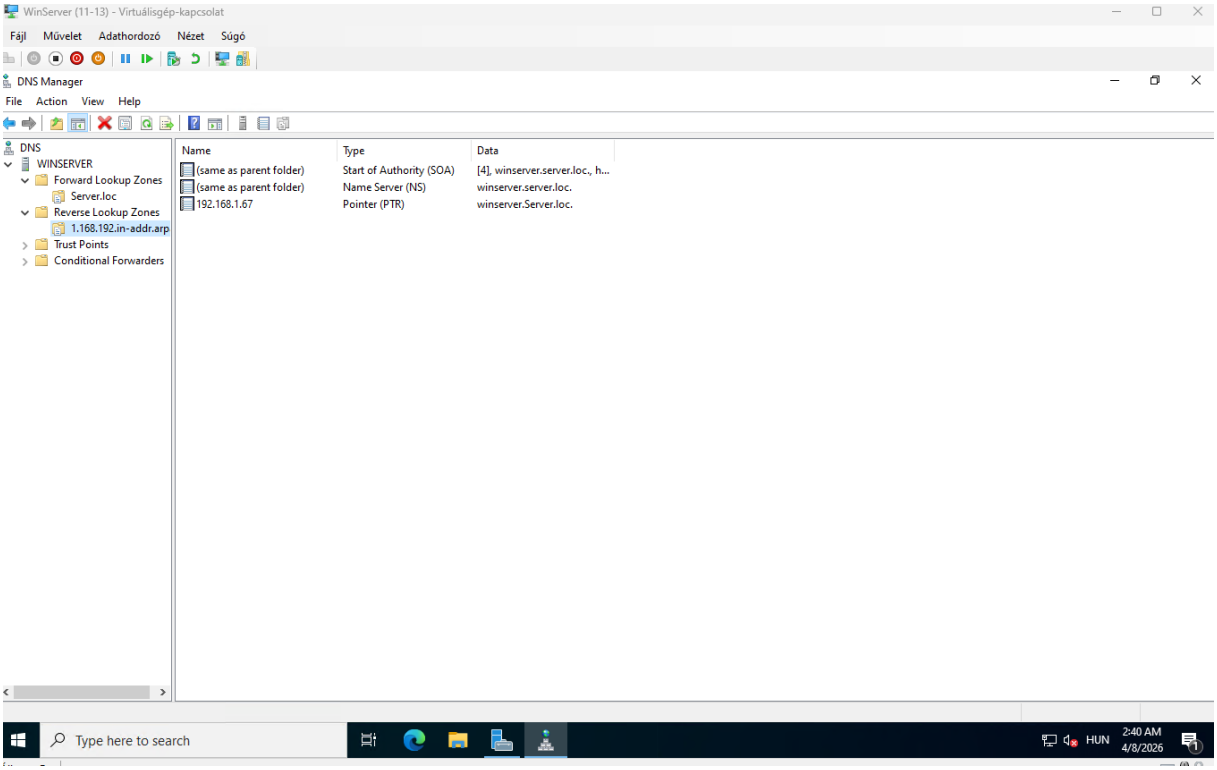
Szerverek:

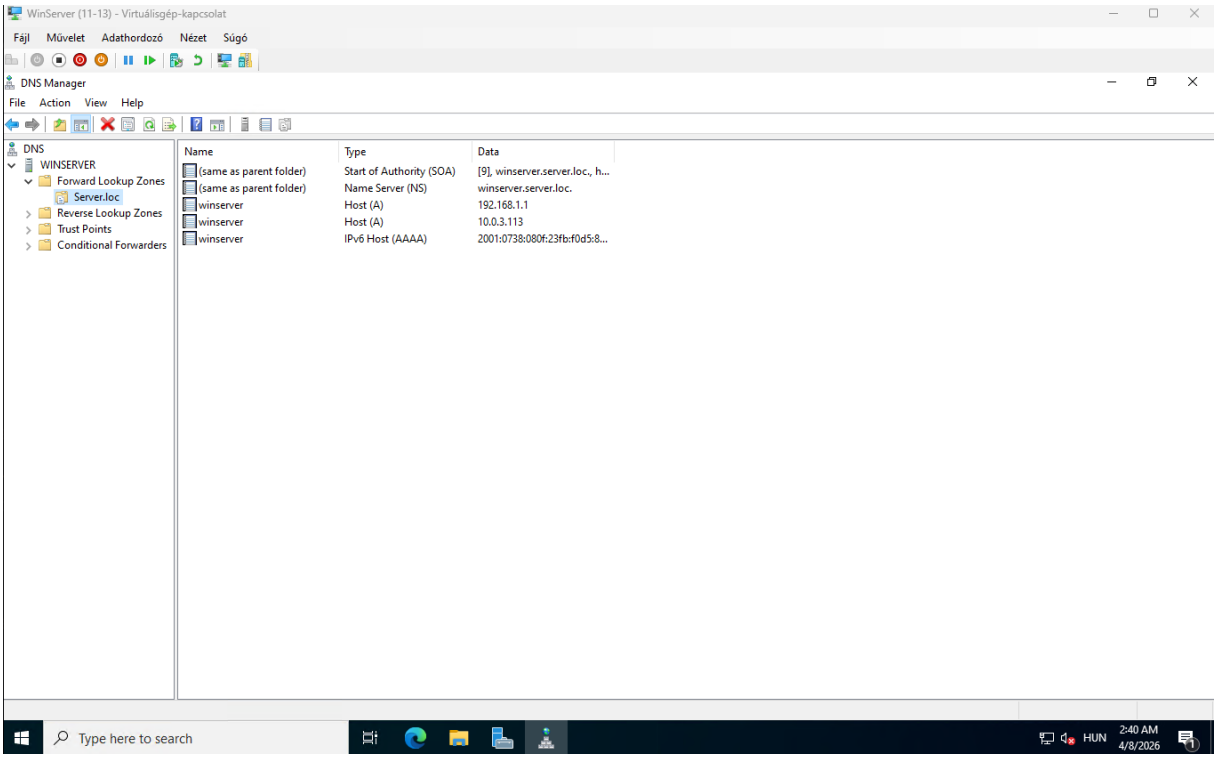
Windows server:

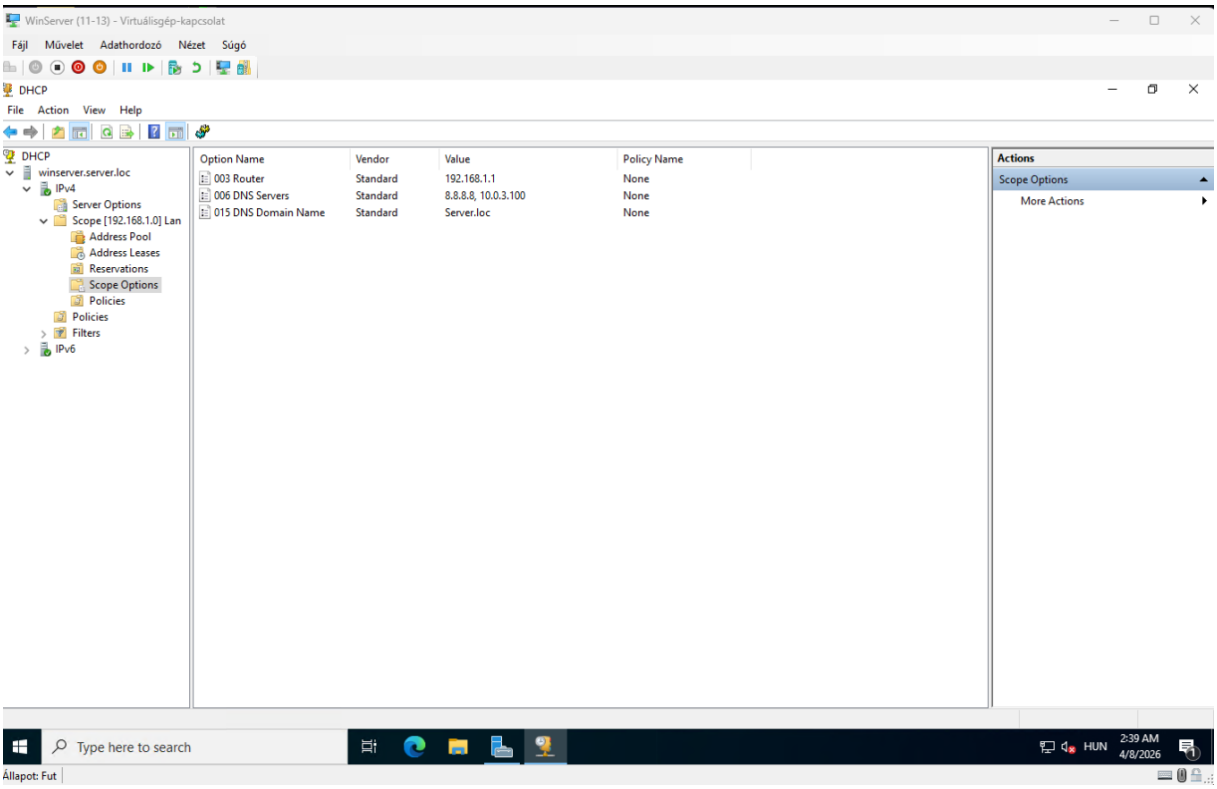

WinServer (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Nézet Súgó
Routing and Remote Access
File Action View Help
Routing and Remote Access
Server Status
WINSERVER (local)
Network Interfaces
Remote Access Logging
IPv4
General
Static Routes
IGMP
NAT
IPv6
IGMP
Interface Protocol State Querier address
LAn Router V3 Querier 192.168.1.1
ÉNet Proxy Up 0.0.0.0
Type here to search
HUN
2:41 AM
4/8/2026
Állapot: Fut

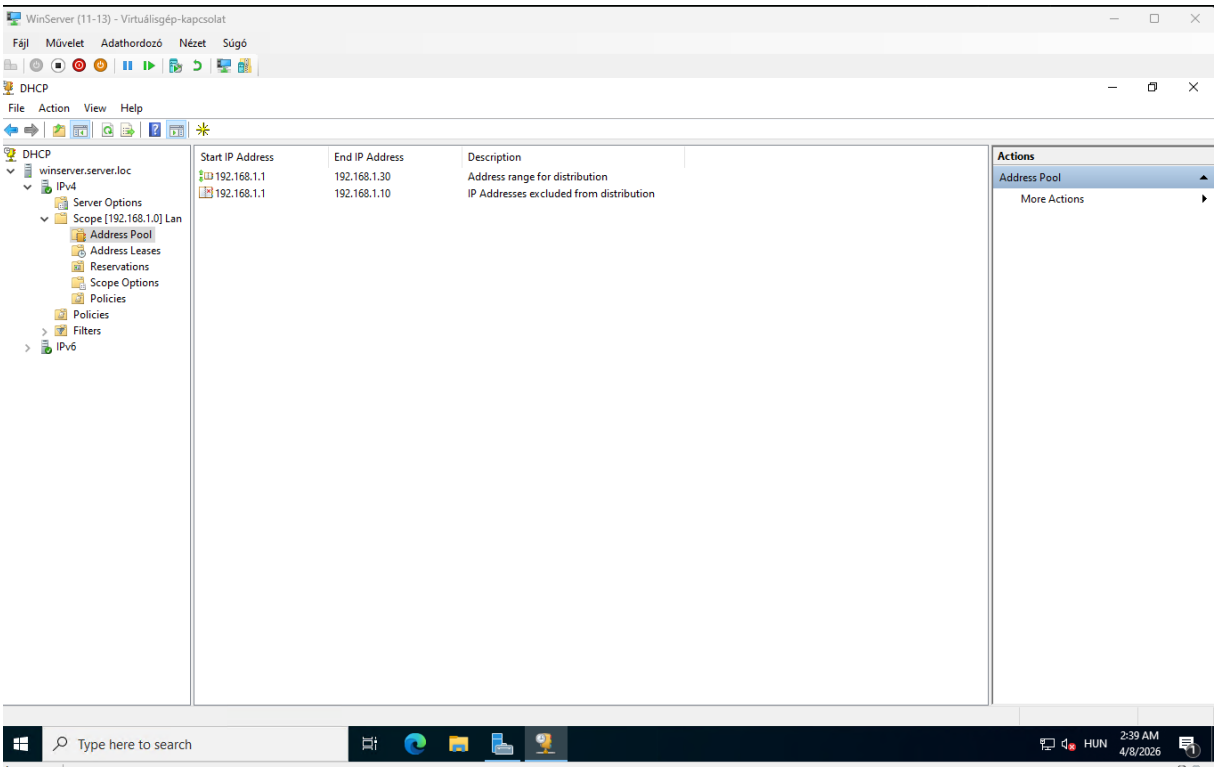

WinServer (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Nézet Súgó
Routing and Remote Access
File Action View Help
Routing and Remote Access
Server Status
WINSERVER (local)
Network Interfaces
Remote Access Logging
IPv4
General
Static Routes
IGMP
NAT
IPv6
NAT
Interface Total mappings Inbound packets translated Inbound packets rejected Outbound packets translated Outbound packets rejected
LAn 0 0 0 0 0
Internal 0 0 0 0 0
ÉNet 26 161,928 0 114,012 0
Type here to search
HUN
2:41 AM
4/8/2026
Állapot: Fut

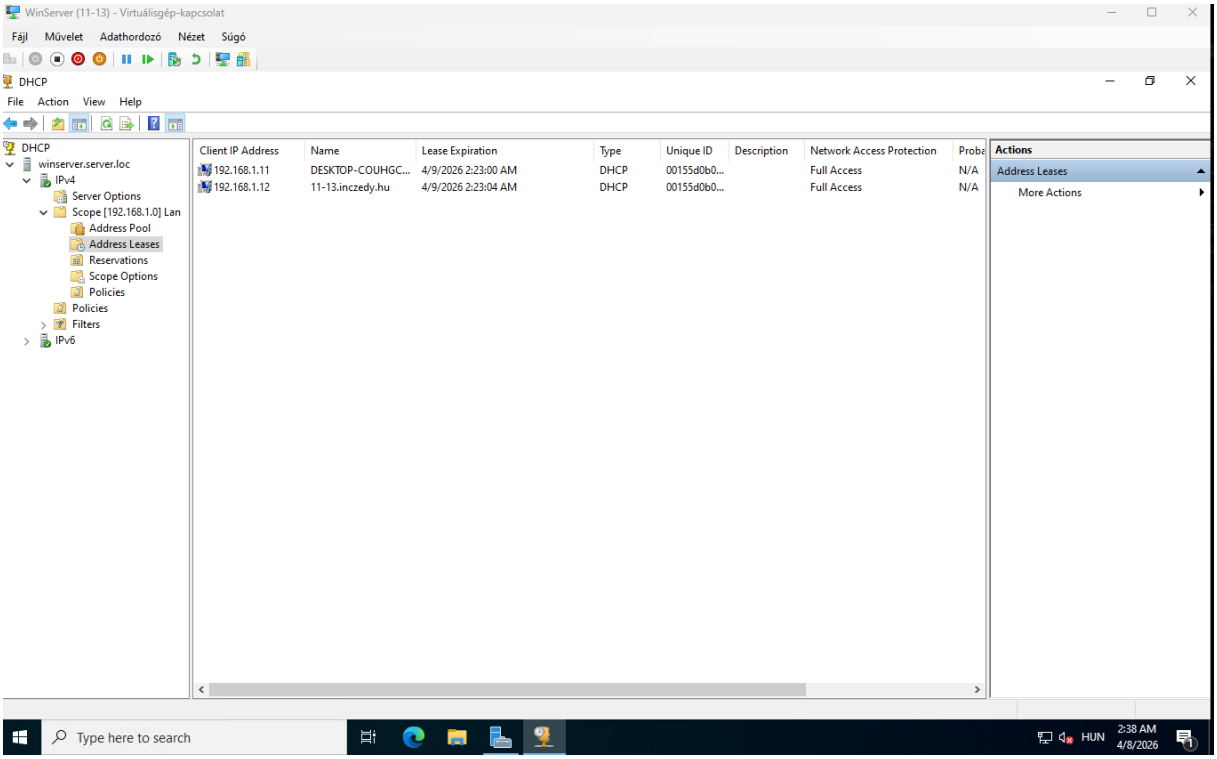

WinServer (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Nézet Súgó
Routing and Remote Access
File Action View Help
Routing and Remote Access
Server Status
WINSERVER (local)
Network Interfaces
Remote Access Logging
IPv4
General
Static Routes
IGMP
NAT
IPv6
General
Interface Type IP Address Incoming bytes Outgoing bytes Static Filters Administrative Status Operational Status
Loopback Loopback 127.0.0.1 0 0 Disabled Up Operational
LAn Dedicated 192.168.1.1 10,959,061 209,011,187 Disabled Up Operational
Internal Internal Not available - - Disabled Unknown Non-operational
ÉNet Dedicated 10.0.3.113 213,153,295 11,595,898 Disabled Up Operational
Type here to search
HUN
2:40 AM
4/8/2026
Állapot: Fut

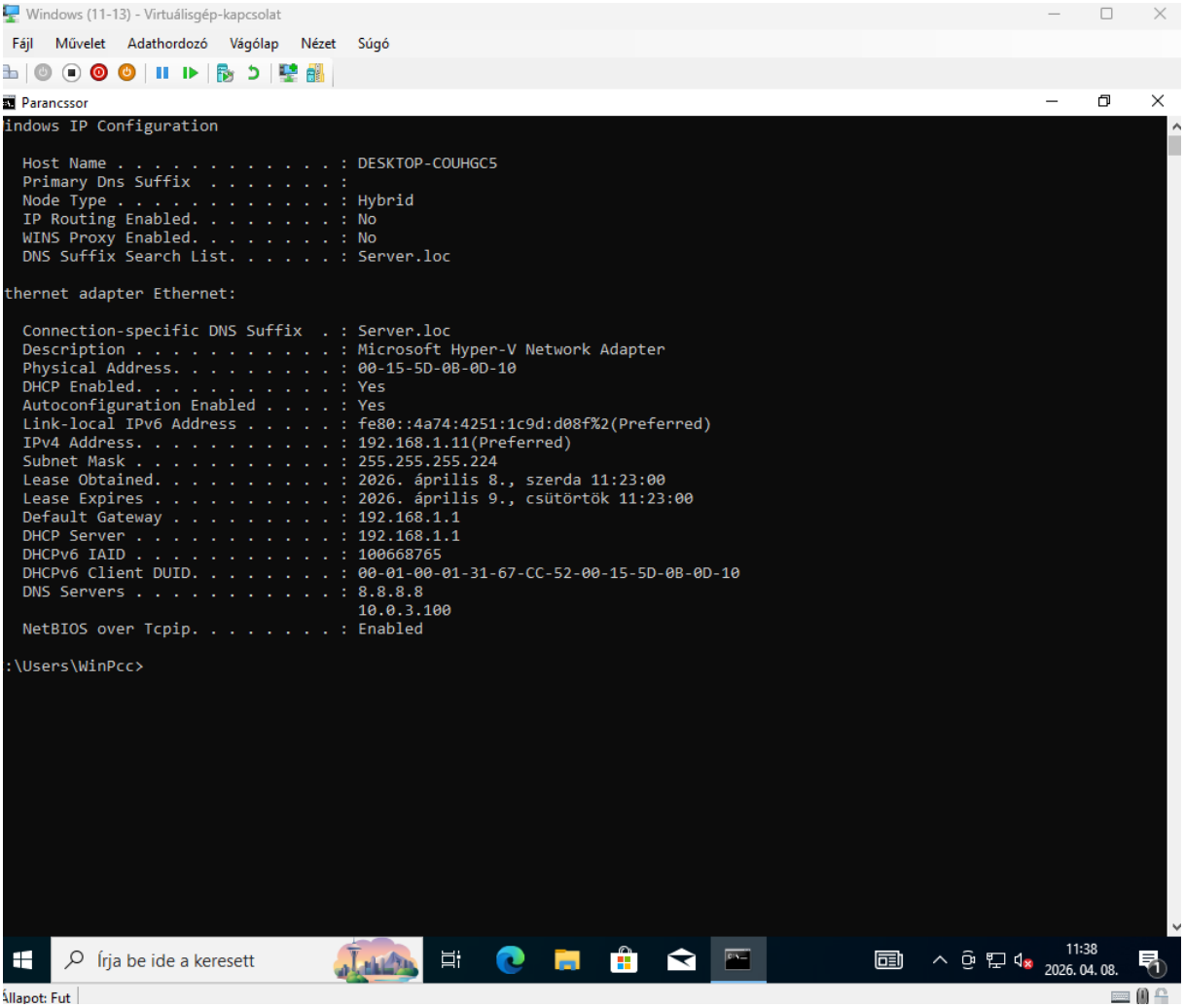

WinServer (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Nézet Súgó
DNS Manager
File Action View Help
DNS
WINSERVER
Forward Lookup Zones
Server.loc
Reverse Lookup Zones
1.168.192.in-addr.arp
Trust Points
Conditional Forwarders
Name Type Data
(same as parent folder) Start of Authority (SOA) [4], winserver.server.loc., h...
(same as parent folder) Name Server (NS) winserver.server.loc.
192.168.1.67 Pointer (PTR) winserver.Server.loc.
Type here to search
HUN
2:40 AM
4/8/2026

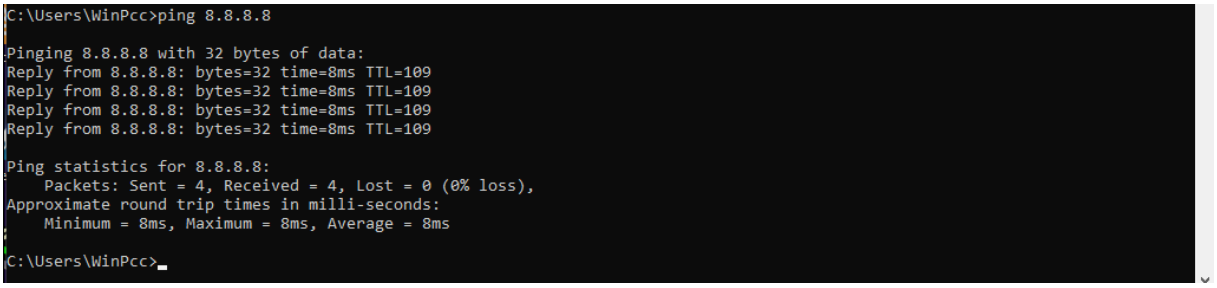

WinServer (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Nézet Súgó
DNS Manager
File Action View Help
DNS
WINSERVER
Forward Lookup Zones
Server.loc
Reverse Lookup Zones
Trust Points
Conditional Forwarders
Name Type Data
(same as parent folder) Start of Authority (SOA) [9], winserver.server.loc., h...
(same as parent folder) Name Server (NS) winserver.server.loc.
winserver Host (A) 192.168.1.1
winserver Host (A) 10.0.3.113
winserver IPv6 Host (AAAA) 2001:0738:080f:23fb:f0d5:8...
Type here to search
HUN 2:40 AM 4/8/2026

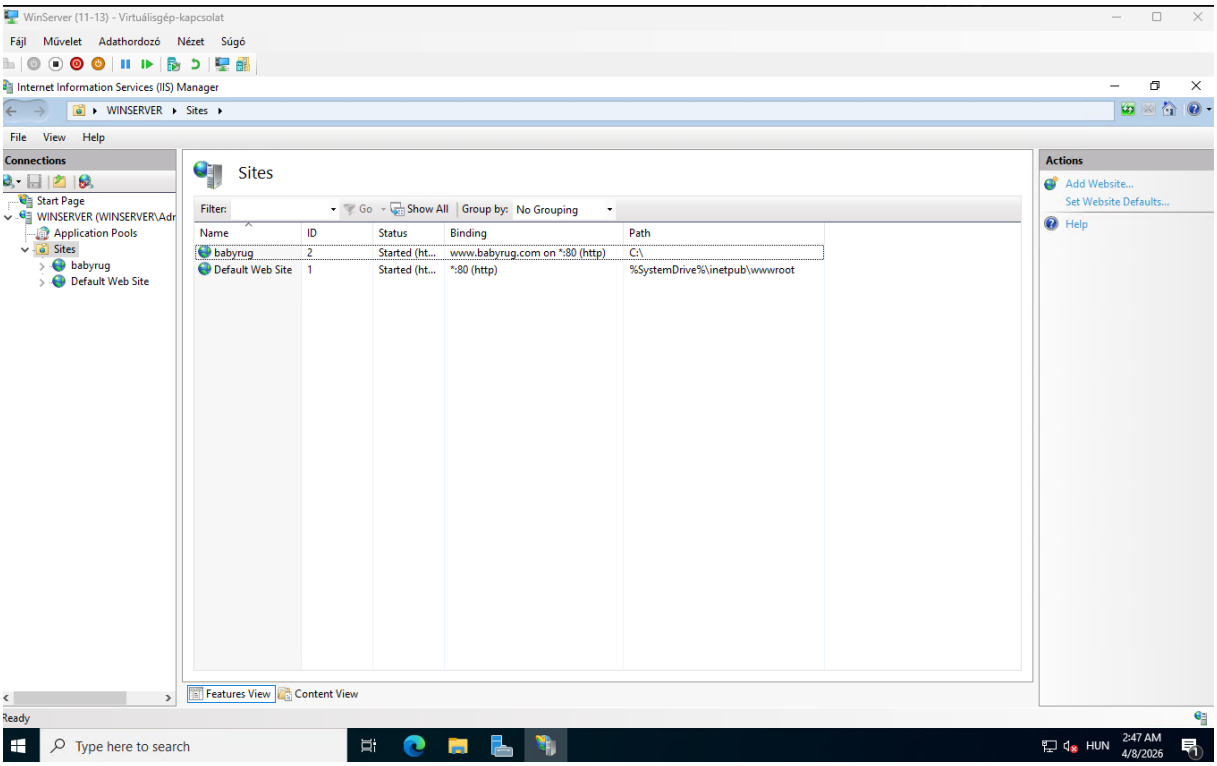

WinServer (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Nézet Súgó
DHCP
File Action View Help
DHCP
winserver.server.loc
IPv4
Server Options
Scope [192.168.1.0] Lan
Address Pool
Address Leases
Reservations
Scope Options
Policies
Policies
Filters
IPv6
Option Name Vendor Value Policy Name
003 Router Standard 192.168.1.1 None
006 DNS Servers Standard 8.8.8.8, 10.0.3.100 None
015 DNS Domain Name Standard Server.loc None
Actions
Scope Options
More Actions
Type here to search
HUN 2:39 AM 4/8/2026
Állapot: Fut

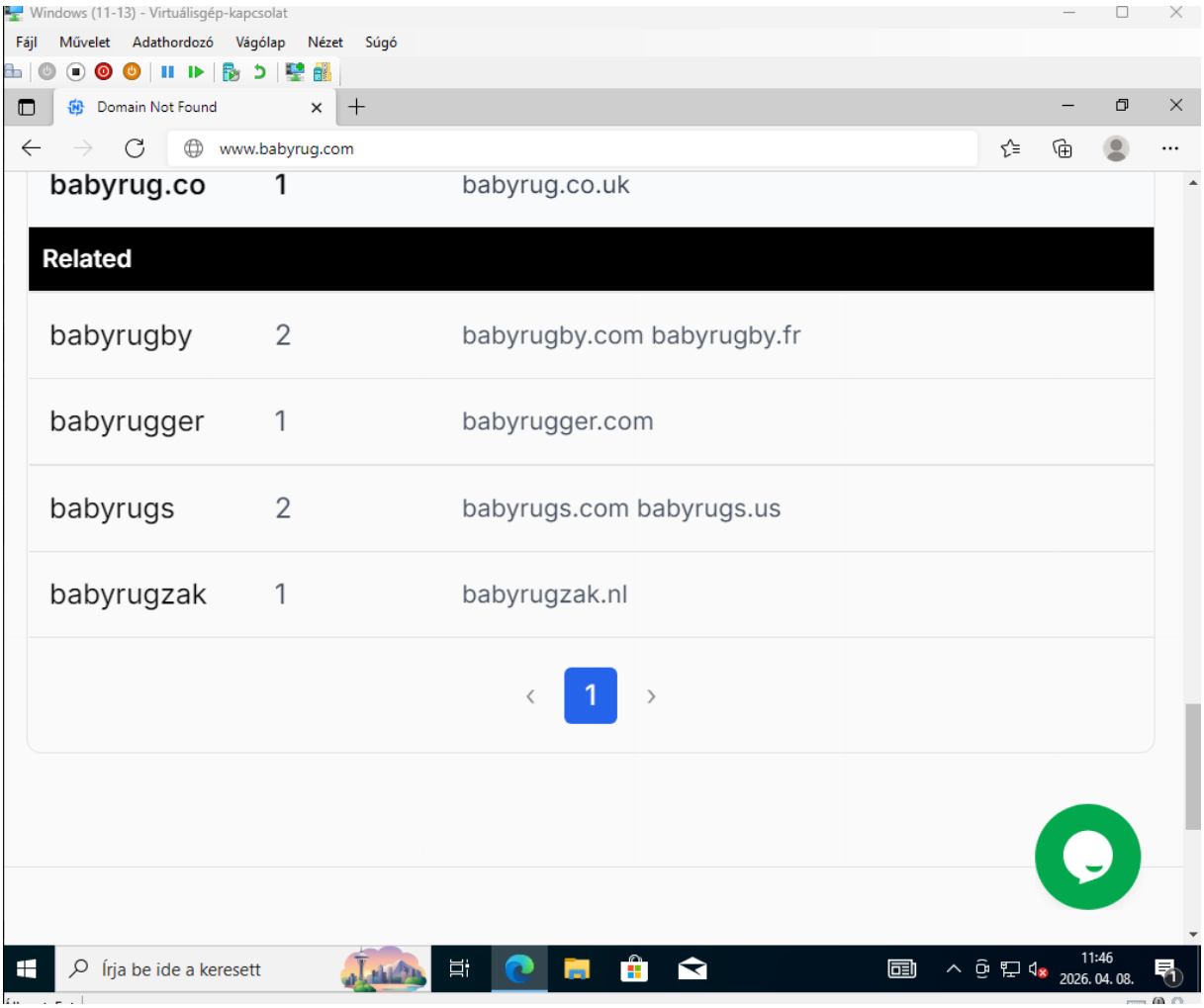

WinServer (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Nézet Súgó
DHCP
File Action View Help
DHCP
winserver.server.loc
IPv4
Server Options
Scope [192.168.1.0] Lan
Address Pool
Address Leases
Reservations
Scope Options
Policies
Policies
Filters
IPv6
Start IP Address
End IP Address
Description
192.168.1.1
192.168.1.30
Address range for distribution
192.168.1.1
192.168.1.10
IP Addresses excluded from distribution
Actions
Address Pool
More Actions
Type here to search
HUN
2:39 AM
4/8/2026


WinServer (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Nézet Súgó
DHCP
File Action View Help
DHCP
winserver.server.loc
IPv4
Server Options
Scope [192.168.1.0] Lan
Address Pool
Address Leases
Reservations
Scope Options
Policies
Policies
Filters
IPv6
Client IP Address
Name
Lease Expiration
Type
Unique ID
Description
Network Access Protection
Proba
192.168.1.11
DESKTOP-COUHGC...
4/9/2026 2:23:00 AM
DHCP
00155d0b0...
Full Access
N/A
192.168.1.12
11-13.inczedy.hu
4/9/2026 2:23:04 AM
DHCP
00155d0b0...
Full Access
N/A
Actions
Address Leases
More Actions
Type here to search
HUN
2:38 AM
4/8/2026


Windows (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Vágólap Nézet Súgó
Parancssor
indows IP Configuration
Host Name . . . . . : DESKTOP-COUHGC5
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : Server.loc
thernet adapter Ethernet:
Connection-specific DNS Suffix . : Server.loc
Description . . . . . : Microsoft Hyper-V Network Adapter
Physical Address. . . . . : 00-15-5D-0B-0D-10
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::4a74:4251:1c9d:d08f%2(Preferred)
IPv4 Address. . . . . : 192.168.1.11(Preferred)
Subnet Mask . . . . . : 255.255.255.224
Lease Obtained. . . . . : 2026. április 8., szerda 11:23:00
Lease Expires . . . . . : 2026. április 9., csütörtök 11:23:00
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 100668765
DHCPv6 Client DUID. . . . . : 00-01-00-01-31-67-CC-52-00-15-5D-0B-0D-10
DNS Servers . . . . . : 8.8.8.8
10.0.3.100
NetBIOS over Tcpip. . . . . : Enabled
:\Users\WinPcc>
Írja be ide a keresett
11:38
2026. 04. 08.
Állapot: Fut


C:\Users\WinPcc>ping 8.8.8.8
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=8ms TTL=109
Reply from 8.8.8.8: bytes=32 time=8ms TTL=109
Reply from 8.8.8.8: bytes=32 time=8ms TTL=109
Reply from 8.8.8.8: bytes=32 time=8ms TTL=109
Ping statistics for 8.8.8.8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 8ms, Maximum = 8ms, Average = 8ms
C:\Users\WinPcc>


WinServer (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Nézet Súgó
Internet Information Services (IIS) Manager
WINSERVER ▸ Sites ▸
File View Help
Connections
Start Page
WINSERVER (WINSERVER\Adr
Application Pools
Sites
babyrug
Default Web Site
Sites
Filter: Go Show All Group by: No Grouping
Name ID Status Binding Path
babyrug 2 Started (ht... www.babyrug.com on *:80 (http) C:\
Default Web Site 1 Started (ht... *:80 (http) %SystemDrive%\inetpub\wwwroot
Features View Content View
Actions
Add Website...
Set Website Defaults...
Help
Ready
Type here to search
HUN
2:47 AM
4/8/2026


Windows (11-13) - Virtuálisgép-kapcsolat
Fájl Művelet Adathordozó Vágólap Nézet Súgó
Domain Not Found
www.babyrug.com
babyrug.co 1 babyrug.co.uk
Related
babyrugby 2 babyrugby.com babyrugby.fr
babyrugger 1 babyrugger.com
babyrugs 2 babyrugs.com babyrugs.us
babyrugzak 1 babyrugzak.nl
1
Írja be ide a keresett
11:46
2026. 04. 08.

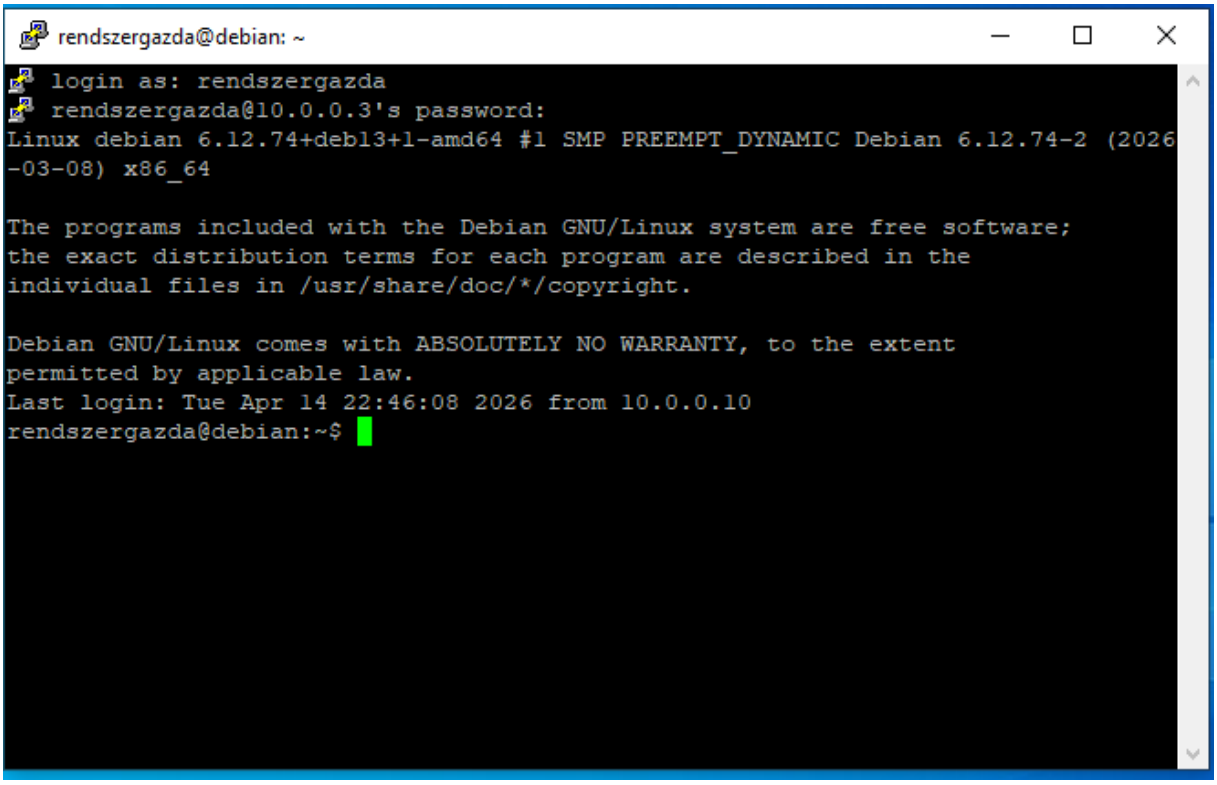
LINUX SZERVER:

ip cím:

```
eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
 link/ether 00:15:5d:01:59:02 brd ff:ff:ff:ff:ff:ff
 altname enx00155d015902
 inet 10.0.0.3/28 brd 10.0.0.15 scope global eth1
    valid_lft forever preferred_lft forever
 inet6 fe80::215:5dff:fe01:5902/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
```

ssh beállítások:

```
Port 22

PermitRootLogin no

PasswordAuthentication yes
```



ufw beállítások:

```
rendszergazda@debian:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
53                         ALLOW IN    192.168.1.64/28
80/tcp (Apache)            ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)
80/tcp (Apache (v6))       ALLOW IN    Anywhere (v6)
```

dns beállítások:

```
net.ipv4.ip_forward=1
```

```
  GNU nano 8.4
options {
        directory "/var/cache/bind";

        recursion yes;
        allow-query { any; };

        listen-on {
                10.0.0.3;
        };

        forwarders {
                8.8.8.8;
                1.1.1.1;
        };

        dnssec-validation auto;

};
```

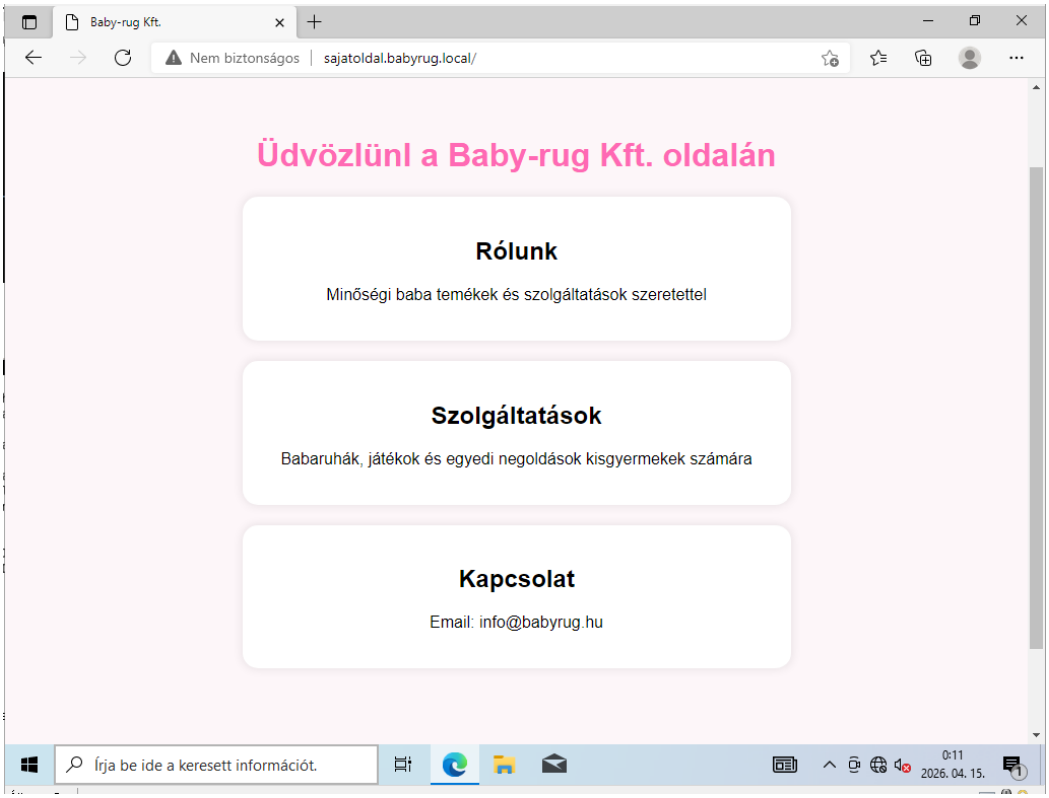
```
  GNU nano 8.4
//
// Do any local configuration here
//

zone "babyrug.local" {
        type master;
        file "/etc/bind/db.babyrug.local";
};
```

```
  GNU nano 8.4                                  /etc/bind/db.babyrug.local
$TTL604800
@ IN SOA debian.babyrug.local. admin.babyrug.local. (
         2026010101
         604800
         86400
         2419200
         604800 )

@ IN NS debian.babyrug.local.
debian IN A 10.0.0.3
kliens IN A 192.168.1.67
sajatoldal IN A 10.0.0.3
```

```
C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : babyrug.local
   Link-local IPv6 Address . . . . . : fe80::a5b0:8751:a5ea:ac85%12
   IPv4 Address. . . . . . . . . . . : 10.0.0.10
   Subnet Mask . . . . . . . . . . . : 255.255.255.240
   Default Gateway . . . . . . . . . : 10.0.0.3
```

apache beállításai:

```
  GNU nano 8.4                    /etc/apache2/sites-available/sajatoldal.conf
<VirtualHost *:80>
        ServerAdmin admin@localhost
        ServerName sajatoldal.babyrug.local

        DocumentRoot /var/www/sajatoldal

        <Directory /var/www/sajatoldal>
                AllowOverride All
                Require all granted
        </Directory>

        ErrorLog ${APACHE_LOG_DIR}/sajatoldal_error.log
        CustomLog ${APACHE_LOG_DIR}/sajatoldal_access.log combined
</VirtualHost>
```



HÁLÓZAT PROGRAMOZÁS:

```
(venv) C:\Users\jocó\Desktop\programozas\szotagszamlalo>pip install paramiko
Collecting paramiko
  Downloading paramiko-4.0.0-py3-none-any.whl.metadata (3.9 kB)
Collecting bcrypt>=3.2 (from paramiko)
  Downloading bcrypt-5.0.0-cp39-abi3-win_amd64.whl.metadata (10 kB)
Collecting cryptography>=3.3 (from paramiko)
  Downloading cryptography-46.0.7-cp311-abi3-win_amd64.whl.metadata (5.7 kB)
Collecting invoke>=2.0 (from paramiko)
  Downloading invoke-3.0.3-py3-none-any.whl.metadata (3.2 kB)
Collecting pynacl>=1.5 (from paramiko)
  Downloading pynacl-1.6.2-cp38-abi3-win_amd64.whl.metadata (10 kB)
Collecting cffi>=2.0.0 (from cryptography>=3.3->paramiko)
  Downloading cffi-2.0.0-cp313-cp313-win_amd64.whl.metadata (2.6 kB)
Collecting pycparser (from cffi>=2.0.0->cryptography>=3.3->paramiko)
  Downloading pycparser-3.0-py3-none-any.whl.metadata (8.2 kB)
Downloading paramiko-4.0.0-py3-none-any.whl (223 kB)
Downloading bcrypt-5.0.0-cp39-abi3-win_amd64.whl (150 kB)
Downloading cryptography-46.0.7-cp311-abi3-win_amd64.whl (3.5 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 3.5/3.5 MB 22.6 MB/s  0:00:00
Downloading cffi-2.0.0-cp313-cp313-win_amd64.whl (183 kB)
Downloading invoke-3.0.3-py3-none-any.whl (160 kB)
Downloading pynacl-1.6.2-cp38-abi3-win_amd64.whl (239 kB)
Downloading pycparser-3.0-py3-none-any.whl (48 kB)
Installing collected packages: pycparser, invoke, bcrypt, cffi, pynacl, cryptography, paramiko
Successfully installed bcrypt-5.0.0 cffi-2.0.0 cryptography-46.0.7 invoke-3.0.3 paramiko-4.0.0 pycparser-3.0 pynacl-1.6.2
```

```python
import paramiko

def get_apache_logs(host, port, username, password, log_path="/var/log/apache2/access.log"):
    try:
        # SSH kliens létrehozása
        client = paramiko.SSHClient()
        client.set_missing_host_key_policy(paramiko.AutoAddPolicy())

        # Kapcsolódás
        client.connect(hostname=host, port=port, username=username, password=password)

        # Parancs futtatása (log lekérés)
        command = f"tail -n 50 {log_path}"
        stdin, stdout, stderr = client.exec_command(command)

        # Kimenet beolvasása
        logs = stdout.read().decode()
        errors = stderr.read().decode()

        client.close()

        if errors:
            print("Hiba történt:")
            print(errors)
        else:
            print("Apache logok (utolsó 50 sor):")
            print(logs)

    except Exception as e:
        print(f"Hiba: {e}")


# Használat
if __name__ == "__main__":
    get_apache_logs(
        host="192.168.1.3",
        port=22,
        username="admin",
        password="cisco123",
        log_path="/var/log/apache2/access.log"
    )
```